Version: 1.1
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Purpose: This document provides a standardized procedure for automating bulk user provisioning in an on-premises Active Directory (AD) environment using PowerShell from a CSV file. It ensures efficiency, reduces errors, and aligns with security best practices for enterprise onboarding. Tested in a VirtualBox lab with Windows Server 2022 DC and Windows 11 workstation.

Scope

- Applies to Windows Server 2022 AD DS environments (lab or production).
- Handles bulk creation of 20+ users with attributes like name, email, department, OU placement, and forced password change.
- Includes troubleshooting for common issues (e.g., OU path syntax, password policy).
- Excludes deprovisioning, hybrid Azure AD sync, or advanced features like email notifications.

Assumptions and Prerequisites

- Domain admin privileges (e.g., lab\Administrator).
- RSAT tools installed on DC (Server Manager > Add Roles > Remote Server Administration Tools > AD Module for PowerShell).
- CSV file with headers: FirstName,LastName,Username,Password,OU,Email,Department.
- Passwords comply with domain policy (e.g., complexity enabled; relaxed in lab if needed).
- OUs created beforehand (IT, HR, Sales, Marketing, Finance).
- Lab setup: VirtualBox with DC VM (IP 192.168.1.10) and optional WS1 VM (IP 192.168.1.20) for testing.
- Execution on the DC or management workstation.

Risks and Mitigations

Risk: Invalid OU paths cause "no superior reference" errors. Mitigation: Validate DN format (no spaces after commas, e.g., OU=IT,DC=lab,DC=local); test manually with New-ADUser.

Risk: Password policy violations (e.g., "too weak"). Mitigation: Temporarily relax Default Domain Policy for lab (Group Policy Management > Edit > Password Policy > Disable complexity); enforce in production.
Risk: Duplicate users. Mitigation: Script checks existence and skips.
Risk: Exposed passwords in CSV. Mitigation: Use secure storage or generate randomly in script; force change at logon.
Compliance: Aligns with NIST SP 800-63B (identity proofing) and ISO 27001 (access control). Flag "-ChangePasswordAtLogon $true" enhances security.

Procedure

Prepare Input Data:

- Create/edit NewUsers.csv in C:\Scripts with validated data (no extra spaces in OU paths).
- Example row: Alice,Johnson,ajohnson,SecurePass2026!,OU=IT,DC=lab,DC=local,ajohnson@lab.local,IT

Script Deployment:

- Save BulkUserEnrollment.ps1 in C:\Scripts (see Appendix A).
- Import module: Import-Module ActiveDirectory.

Execution:

- Run as domain admin: cd C:\Scripts > .\BulkUserEnrollment.ps1.
- Monitor for green success messages.

Verification:

- Refresh AD Users and Computers; check OUs for users.
- Query: Get-ADUser -Filter * -Properties EmailAddress, Department.
- Test login on domain-joined WS1 (e.g., lab\ajohnson).
- Audit: Event Viewer > Security > Event ID 4720 (user created).

Monitoring and Maintenance

- Log runs: Add Start-Transcript -Path C:\Logs\Enrollment.log to script.
- Schedule via Task Scheduler for HR integrations.
- Update for new attributes (e.g., add -Title, -OfficePhone).

Appendix A: PowerShell Script

```
# Import Active Directory module
Import-Module ActiveDirectory

# Path to CSV file
$csvPath = "C:\Scripts\NewUsers.csv"

# Import user data from CSV
$users = Import-Csv -Path $csvPath

# Loop through each user in the CSV
foreach ($user in $users) {
    $firstName = $user.FirstName
    $lastName = $user.LastName
    $username = $user.Username
    $password = $user.Password | ConvertTo-SecureString -AsPlainText -Force
    $ou = $user.OU
    $email = $user.Email
    $department = $user.Department

    # Full name and display name
    $fullName = "$firstName $lastName"

    # Check if user already exists
    if (Get-ADUser -Filter {SamAccountName -eq $username} -ErrorAction
SilentlyContinue) {
        Write-Host "User $username already exists. Skipping." -ForegroundColor
Yellow
        continue
    }

    # Create new AD user
    try {
        New-ADUser `
            -Name $fullName `
            -GivenName $firstName `
            -Surname $lastName `
            -SamAccountName $username `
```

```
            -UserPrincipalName "$username@lab.local" `
            -Path "$ou" `
            -AccountPassword $password `
            -EmailAddress $email `
            -Department $department `
            -Enabled $true `
            -ChangePasswordAtLogon $true  # Force password change on first login

        Write-Host "Created user: $username" -ForegroundColor Green

    } catch {
        Write-Host "Error creating user ${username}: $_" -ForegroundColor Red
    }
}

Write-Host "Bulk user enrollment complete!" -ForegroundColor Cyan
```

Appendix B: Sample CSV

```
FirstName,LastName,Username,Password,OU,Email,Department
Alice,Johnson,ajohnson,SecurePass2026!,OU=IT,DC=lab,DC=local,ajohnson@lab.local,IT
Bob,Smith,bsmith,SecurePass2026!,OU=HR,DC=lab,DC=local,bsmith@lab.local,HR
Charlie,Davis,cdavis,SecurePass2026!,OU=Sales,DC=lab,DC=local,cdavis@lab.local,Sales
Dana,Lee,dlee,SecurePass2026!,OU=Marketing,DC=lab,DC=local,dlee@lab.local,Marketing
Evan,Miller,emiller,SecurePass2026!,OU=Finance,DC=lab,DC=local,emiller@lab.local,Finance
Fiona,Wilson,fwilson,SecurePass2026!,OU=IT,DC=lab,DC=local,fwilson@lab.local,IT
George,Harris,gharris,SecurePass2026!,OU=HR,DC=lab,DC=local,gharris@lab.local,HR
Hannah,Clark,hclark,SecurePass2026!,OU=Sales,DC=lab,DC=local,hclark@lab.local,Sales
Ian,Robinson,irobinson,SecurePass2026!,OU=Marketing,DC=lab,DC=local,irobinson@lab.local,Marketing
```

Jenna,Lewis,jlewis,SecurePass2026!,OU=Finance,DC=lab,DC=local,jlewis@lab.local,Finance
Kevin,Walker,kwalker,SecurePass2026!,OU=IT,DC=lab,DC=local,kwalker@lab.local,IT
Laura,Hall,lhall,SecurePass2026!,OU=HR,DC=lab,DC=local,lhall@lab.local,HR
Mike,Allen,mallen,SecurePass2026!,OU=Sales,DC=lab,DC=local,mallen@lab.local,Sales
Nina,Young,nyoung,SecurePass2026!,OU=Marketing,DC=lab,DC=local,nyoung@lab.local,Marketing
Oscar,King,oking,SecurePass2026!,OU=Finance,DC=lab,DC=local,oking@lab.local,Finance
Paula,Wright,pwright,SecurePass2026!,OU=IT,DC=lab,DC=local,pwright@lab.local,IT
Quinn,Scott,qscott,SecurePass2026!,OU=HR,DC=lab,DC=local,qscott@lab.local,HR
Riley,Green,rgreen,SecurePass2026!,OU=Sales,DC=lab,DC=local,rgreen@lab.local,Sales
Sam,Torres,storres,SecurePass2026!,OU=Marketing,DC=lab,DC=local,storres@lab.local,Marketing
Tina,Adams,tadams,SecurePass2026!,OU=Finance,DC=lab,DC=local,tadams@lab.local,Finance

Appendix C: Screenshots and Learnings

Lab1-BulkEnrollment [Running] - Oracle VirtualBox

File Machine View Input Devices Help

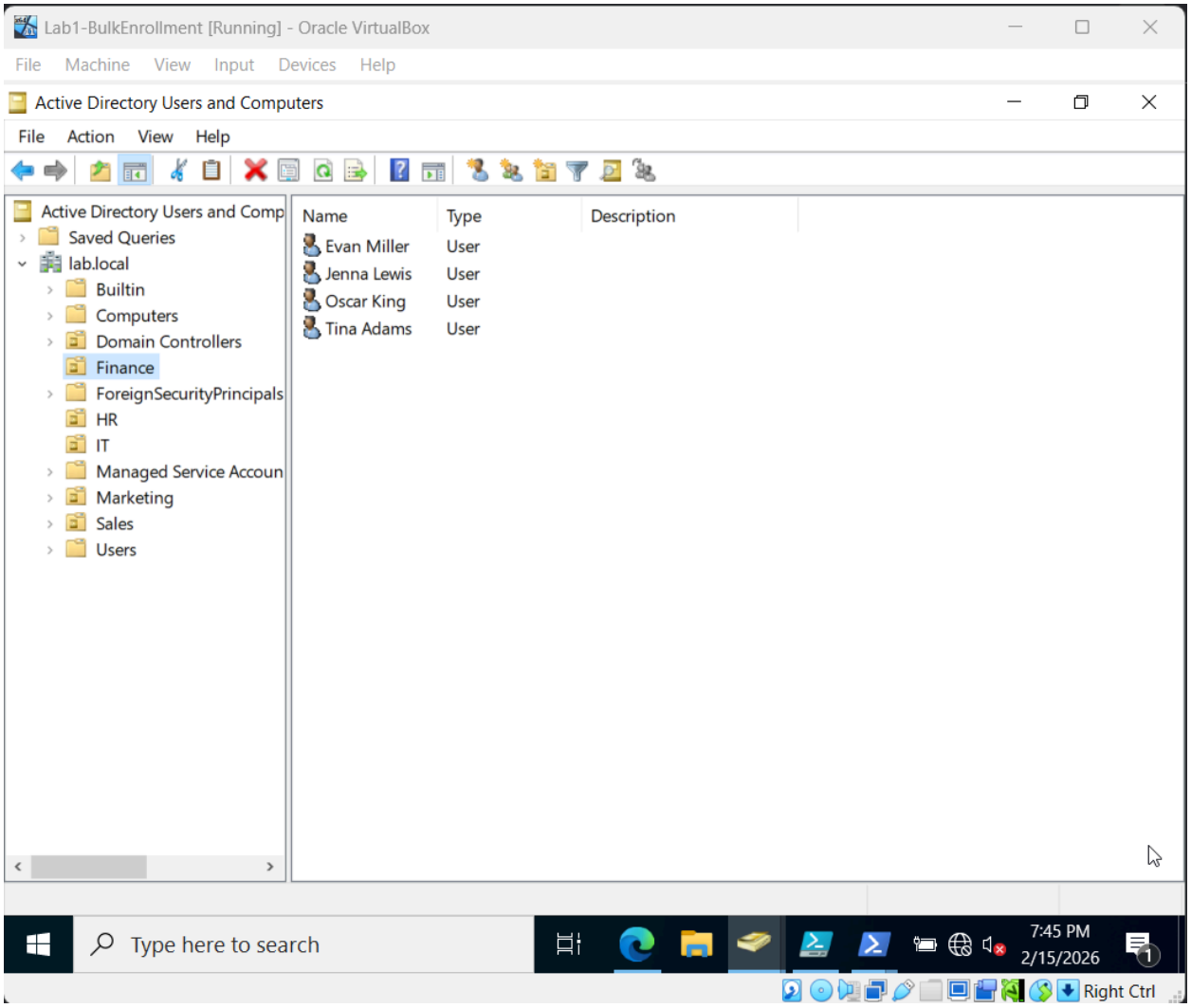
Administrator: Windows PowerShell

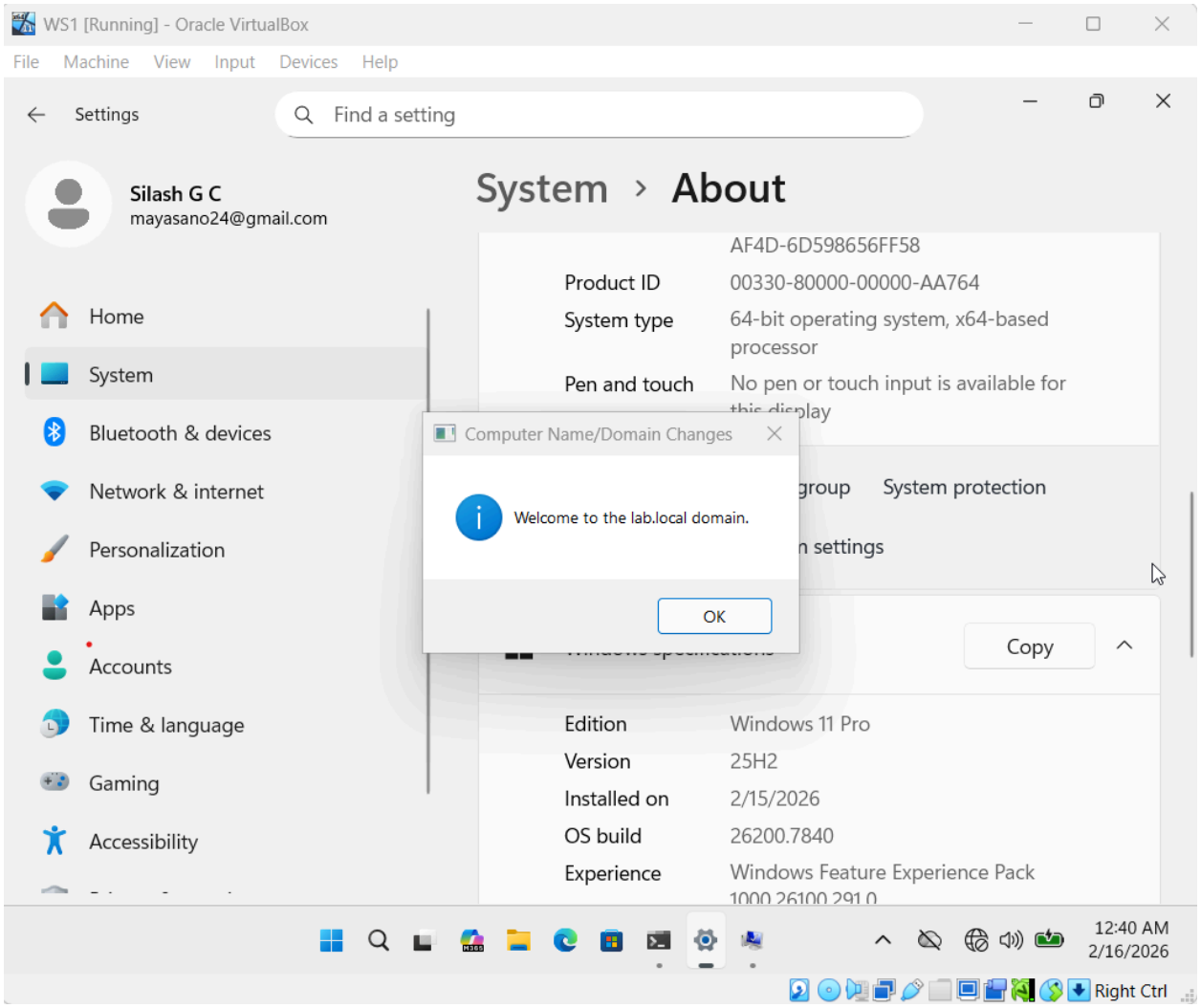
```
PS C:\Scripts> .\BulkUserEnrollment.ps1
Created user: ajohnson
Created user: bsmith
Created user: cdavis
Created user: dlee
Created user: emiller
Created user: fwilson
Created user: gharris
Created user: hclark
Created user: irobinson
Created user: jlewis
Created user: kwalker
Created user: lhall
Created user: mallen
Created user: nyoung
Created user: oking
Created user: pwright
Created user: qscott
Created user: rgreen
Created user: storres
Created user: tadams
Bulk user enrollment complete!
PS C:\Scripts>
```

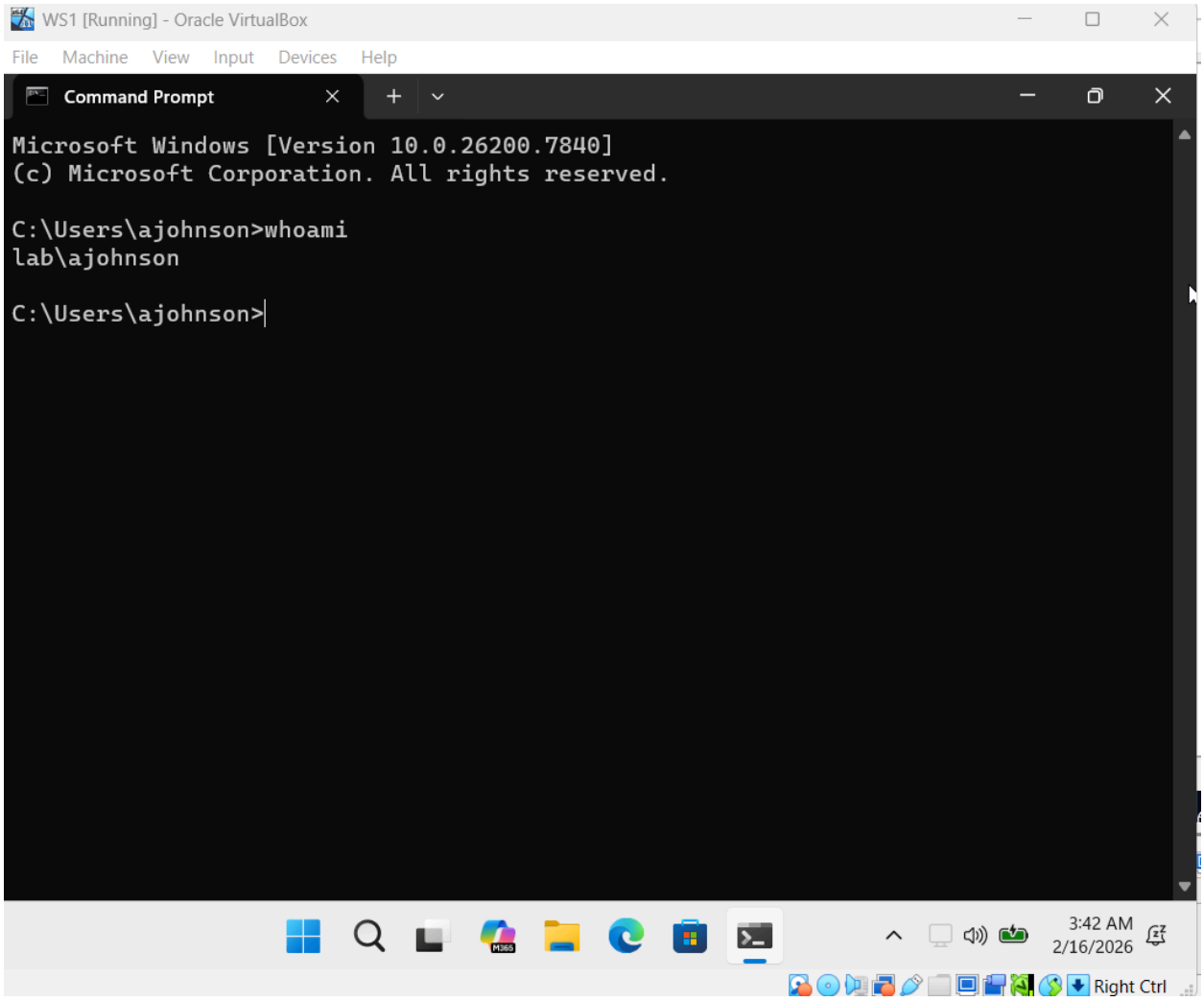
Type here to search

7:40 PM
2/15/2026

Right Ctrl


Lab1-BulkEnrollment [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Active Directory Users and Computers
File Action View Help
Active Directory Users and Comp
Saved Queries
lab.local
Builtin
Computers
Domain Controllers
Finance
ForeignSecurityPrincipals
HR
IT
Managed Service Accoun
Marketing
Sales
Users
Name Type Description
Evan Miller User
Jenna Lewis User
Oscar King User
Tina Adams User
Type here to search
7:45 PM
2/15/2026
Right Ctrl


WS1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Settings
Find a setting
Silash G C
mayasano24@gmail.com
System › About
Home
System
Bluetooth & devices
Network & internet
Personalization
Apps
Accounts
Time & language
Gaming
Accessibility
AF4D-6D598656FF58
Product ID 00330-80000-00000-AA764
System type 64-bit operating system, x64-based processor
Pen and touch No pen or touch input is available for this display
Computer Name/Domain Changes
Welcome to the lab.local domain.
OK
group
System protection
n settings
Copy
Edition Windows 11 Pro
Version 25H2
Installed on 2/15/2026
OS build 26200.7840
Experience Windows Feature Experience Pack
12:40 AM
2/16/2026
Right Ctrl



Key Learning: Ensure OU paths have no spaces after commas to avoid syntax errors in New-ADUser.